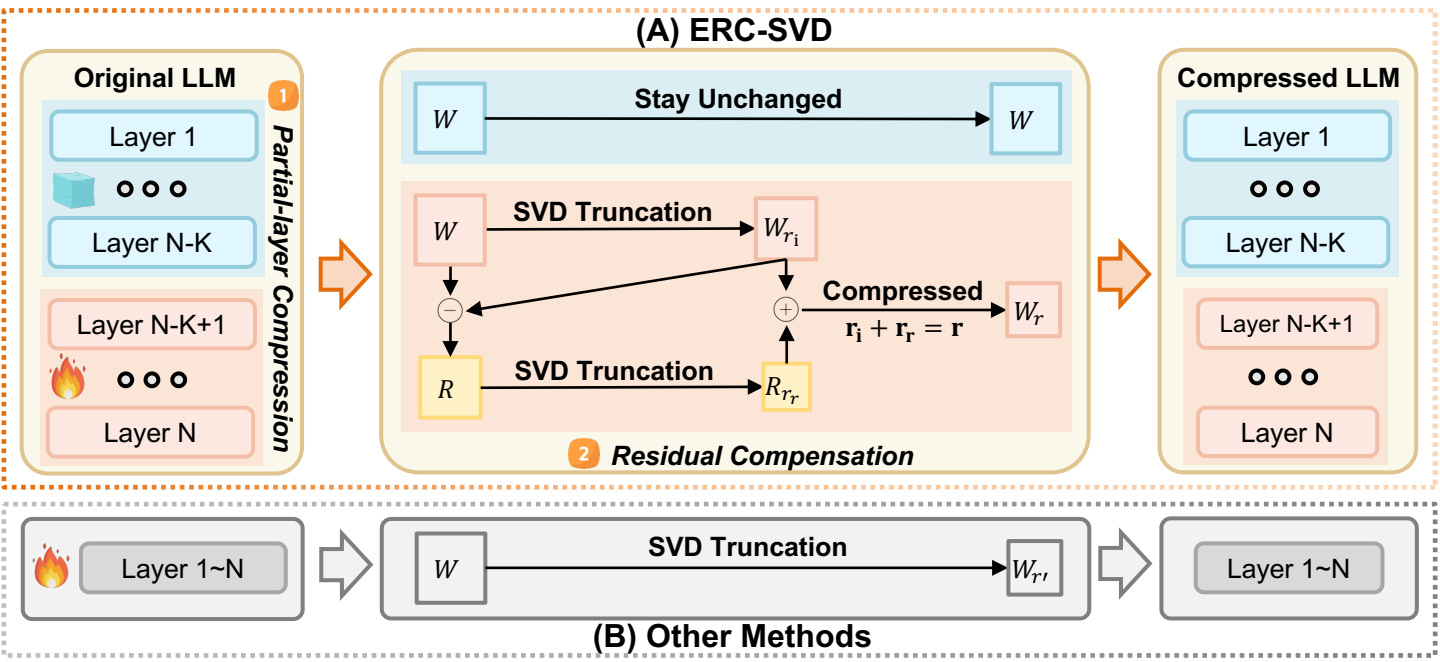

(A) ERC-SVD
Original LLM
1
Partial-layer Compression
Layer 1
Layer N-K
Layer N-K+1
Layer N
W
Stay Unchanged
W
SVD Truncation
W_{r_i}
Compressed
$\mathbf{r_i} + \mathbf{r_r} = \mathbf{r}$
W_r
R
SVD Truncation
R_{r_r}
2
Residual Compensation
Compressed LLM
Layer 1
Layer N-K
Layer N-K+1
Layer N
Layer 1~N
W
SVD Truncation
$W_{r'}$
Layer 1~N
(B) Other Methods